Lecture 16: Difference-in-differences

Economics 326 — Introduction to Econometrics II

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Motivation

- In the previous lecture, we estimated treatment effects from **cross-sectional** data under the assumption of selection on observables: after controlling for covariates, treatment is as good as random.
- In many settings, this assumption is hard to justify. An alternative approach exploits **panel data** (repeated observations on the same units over time).
- The **difference-in-differences (DID)** method compares changes over time between a treatment group and a control group.

DID basic setup

- Two time periods: $t \in \{0, 1\}$ (before and after treatment).
- Two groups: $D_i \in \{0, 1\}$ (control and treatment).
- Treatment occurs between periods 0 and 1, and only the treatment group ($D_i = 1$) is affected.
- We observe Y_{it}: the outcome for individual i at time t.

DID regression model

- The DID regression is:

$$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta (t \cdot D_i) + U_{it},$$

  where $\mathrm{E}[U_{it} \mid D_i] = 0$.
- The regressors:
  - t: time indicator (0 = before, 1 = after).
  - D_i: group indicator (0 = control, 1 = treatment).
  - $t \cdot D_i$: interaction term, equals 1 only for the treatment group after treatment.

Interpreting the coefficients

- $$Y_{it} = \alpha + \delta \cdot t + \gamma D_i + \beta (t \cdot D_i) + U_{it},$$
- Evaluate $\mathrm{E}[Y_{it} \mid D_i]$ for each combination of t and D_i:

$$\begin{aligned}
t = 0,\ D_i = 0: &\quad \mathrm{E}[Y_{i0} \mid D_i = 0] = \alpha, \\
t = 0,\ D_i = 1: &\quad \mathrm{E}[Y_{i0} \mid D_i = 1] = \alpha + \gamma, \\
t = 1,\ D_i = 0: &\quad \mathrm{E}[Y_{i1} \mid D_i = 0] = \alpha + \delta, \\
t = 1,\ D_i = 1: &\quad \mathrm{E}[Y_{i1} \mid D_i = 1] = \alpha + \delta + \gamma + \beta.
\end{aligned}$$

- Summarized as a 2×2 table:

	$D_i = 0$ (Control)	$D_i = 1$ (Treatment)
$t = 0$	α	$\alpha + \gamma$
$t = 1$	$\alpha + \delta$	$\alpha + \delta + \gamma + \beta$

- α: baseline expected outcome (control group, before treatment).
- γ: pre-existing **group difference** at baseline ($t = 0$).
- δ: **time effect** — the change in the control group from $t = 0$ to $t = 1$, capturing common trends.
- From the 2×2 table, the change over time for each group:
  - **Treatment:** $\mathrm{E}[Y_{i1} \mid D_i = 1] - \mathrm{E}[Y_{i0} \mid D_i = 1] = \delta + \beta$.
  - **Control:** $\mathrm{E}[Y_{i1} \mid D_i = 0] - \mathrm{E}[Y_{i0} \mid D_i = 0] = \delta$.
- Subtracting the control group's change from the treatment group's change:

$$(\delta + \beta) - \delta = \beta.$$

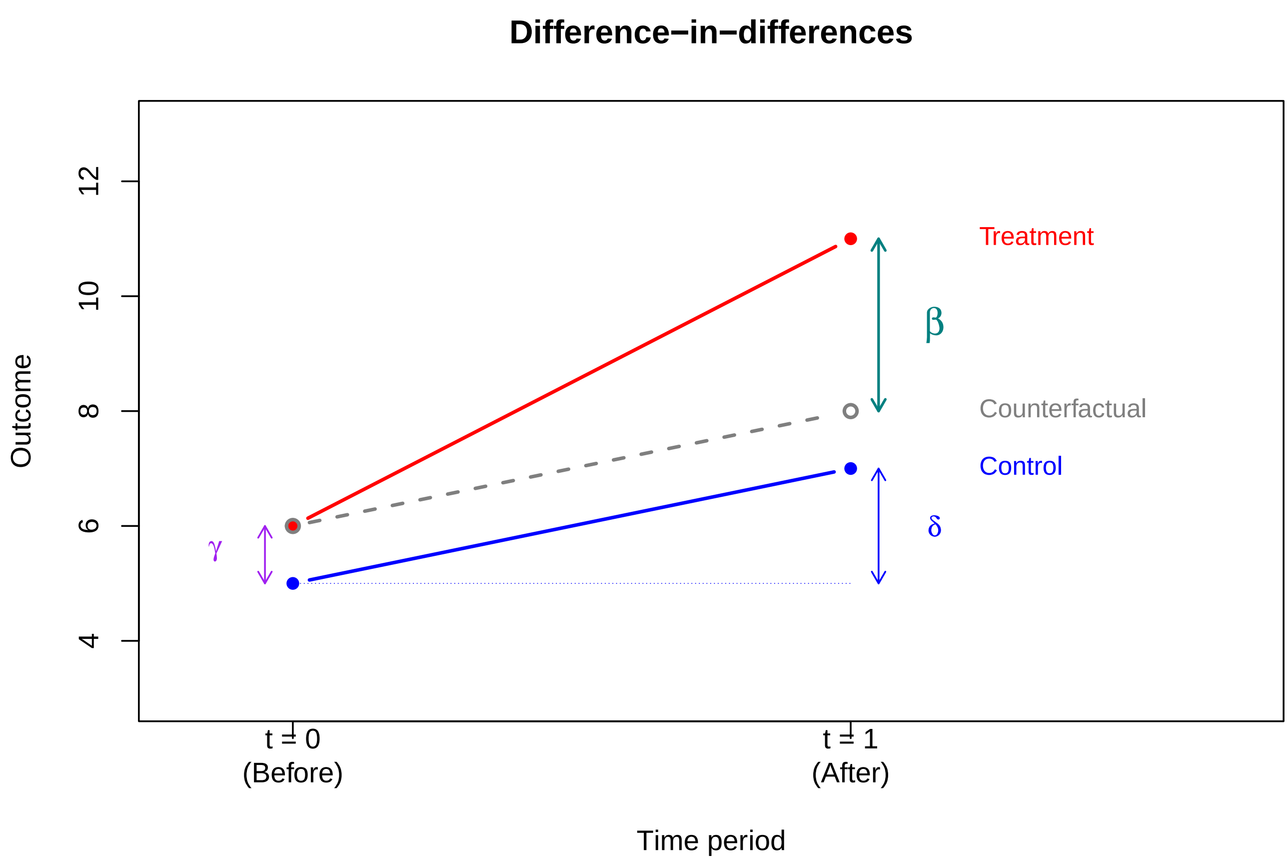
- The DID estimand as a double difference of conditional expectations:

$$\beta = \mathrm{E}[Y_{i1} - Y_{i0} \mid D_i = 1] - \mathrm{E}[Y_{i1} - Y_{i0} \mid D_i = 0].$$

- By subtracting the control group's change, the common time trend δ cancels, isolating the treatment effect β.

DID diagram

- The classic DID diagram shows the control group, treatment group, and counterfactual:



- We predict the counterfactual outcome for the treatment group at $t = 1$ (dashed gray line) by adding the control group's change δ to the treatment group's baseline $\alpha + \gamma$.

DID and potential outcomes

- To connect DID with the potential outcomes framework, define **panel potential outcomes**: $Y_{it}(d)$ is the outcome for individual i at time t if assigned to group $d \in \{0, 1\}$.
- Four potential outcomes: $Y_{i0}(0)$, $Y_{i1}(0)$, $Y_{i0}(1)$, $Y_{i1}(1)$.
- The observed outcome is:

$$Y_{it} = D_i\, Y_{it}(1) + (1 - D_i)\, Y_{it}(0).$$

- What we observe for each group:

	Control ($D_i = 0$)	Treatment ($D_i = 1$)
$t = 0$	$Y_{i0}(0)$	$Y_{i0}(1)$
$t = 1$	$Y_{i1}(0)$	$Y_{i1}(1)$

- The treatment effect at time $t = 1$ for the treated group is:

$$\text{ATT} = \mathrm{E}\left[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1\right].$$

The counterfactual $Y_{i1}(0)$ is unobserved for the treated group.

DID as a treatment effect

- $\beta = \mathrm{E}\left[Y_{i1} - Y_{i0} \mid D_i = 1\right] - \mathrm{E}\left[Y_{i1} - Y_{i0} \mid D_i = 0\right]$.
- Substituting observed outcomes with potential outcomes:

$$\begin{aligned}\beta = {} & \mathrm{E}\left[Y_{i1}(1) - Y_{i0}(1) \mid D_i = 1\right] \\ & - \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right].\end{aligned}$$

- To relate β to the ATT, add and subtract $Y_{i1}(0)$ and $Y_{i0}(0)$ inside the first expectation:

$$\begin{aligned}\beta &= \mathrm{E}\left[Y_{i1}(1) - Y_{i0}(1) \mid D_i = 1\right] - \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right] \\ &= \mathrm{E}\left[Y_{i1}(1) \underbrace{- Y_{i1}(0) + Y_{i1}(0)}_{=0} \underbrace{- Y_{i0}(0) + Y_{i0}(0)}_{=0} - Y_{i0}(1) \mid D_i = 1\right] \\ &\quad - \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right].\end{aligned}$$

- Rearranging and splitting the expectation:

$$\begin{aligned}\beta = {} & \underbrace{\mathrm{E}\left[Y_{i1}(1) - Y_{i1}(0) \mid D_i = 1\right]}_{\text{ATT}} \\ & + \underbrace{\mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1\right] - \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right]}_{\text{difference in trends}} \\ & + \underbrace{\mathrm{E}\left[Y_{i0}(0) - Y_{i0}(1) \mid D_i = 1\right]}_{\text{anticipation effect}}.\end{aligned}$$

- For β to equal the ATT, the difference in trends and anticipation effect must be zero. This requires two assumptions.

Assumption 1: Parallel trends

- The "difference in trends" term equals zero if both groups would have experienced the same change over time in the absence of treatment:

$$\mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1\right] = \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right].$$

- Under parallel trends, the decomposition reduces to:

$$\beta = \mathrm{ATT} + \underbrace{\mathrm{E}\left[Y_{i0}(0) - Y_{i0}(1) \mid D_i = 1\right]}_{\text{anticipation effect}}.$$

- The parallel trends assumption cannot be directly tested because $Y_{i1}(0)$ is unobserved for the treated group.
- If pre-treatment data for multiple periods exist, one can check whether trends were parallel before treatment.

Assumption 2: No anticipation

- The "anticipation effect" term equals zero if the outcome at $t = 0$ (before treatment) is not affected by future treatment assignment:

$$\mathrm{E}\left[Y_{i0}(1) \mid D_i = 1\right] = \mathrm{E}\left[Y_{i0}(0) \mid D_i = 1\right].$$

- Being assigned to the treatment group does not change pre-treatment outcomes in expectation.
- Under both parallel trends and no anticipation:

$$\beta = \mathrm{ATT}.$$

Example: incinerator and house prices

- Kiel and McClain (1995) studied how the construction of a garbage incinerator affected nearby house prices in North Andover, Massachusetts. This is an example of an **event study**: a research design that estimates the causal effect of a specific event by comparing outcomes before and after it occurs.
- We use the `kielmc` dataset from the `wooldridge` package:

```r
library(wooldridge)
data(kielmc)
# Show 2 observations from each of the 4 groups
rows <- c(
  head(which(kielmc$y81 == 0 & kielmc$nearinc == 0), 2),
  head(which(kielmc$y81 == 0 & kielmc$nearinc == 1), 2),
  head(which(kielmc$y81 == 1 & kielmc$nearinc == 0), 2),
  head(which(kielmc$y81 == 1 & kielmc$nearinc == 1), 2)
)
kielmc[rows, c("rprice", "y81", "nearinc", "age")]
```

```
     rprice y81 nearinc age
14   52000.00  0        0  32
15   49000.00  0        0  18
1    60000.00  0        1  48
2    40000.00  0        1  83
187  90245.77  1        0   1
188  46082.95  1        0  41
180  37634.41  1        1  81
181  39938.55  1        1  71
```

- `rprice`: house price in 1978 dollars (Y_{it}).
- `y81`: 1 if year is 1981 (after incinerator announced), 0 if 1978 ($t = 1$ if 1981, 0 if 1978).
- `nearinc`: 1 if house is near the incinerator site (D_i).

The 2×2 table of means

- Compute the four group means:

```r
means <- tapply(kielmc$rprice, list(kielmc$y81, kielmc$nearinc), mean)
colnames(means) <- c("Far (nearinc=0)", "Near (nearinc=1)")
rownames(means) <- c("1978 (y81=0)", "1981 (y81=1)")
round(means, 2)
```

```
             Far (nearinc=0) Near (nearinc=1)
1978 (y81=0)        82517.23         63692.86
1981 (y81=1)       101307.51         70619.24
```

- Computing the DID by hand:

```r
diff_near <- means[2, 2] - means[1, 2]
diff_far  <- means[2, 1] - means[1, 1]
DID <- diff_near - diff_far
cat("Change (near):", round(diff_near, 2), "\n")
```

```
Change (near): 6926.38
```

```r
cat("Change (far): ", round(diff_far, 2), "\n")
```

```
Change (far):  18790.29
```

```r
cat("DID:          ", round(DID, 2), "\n")
```

```
DID:           -11863.9
```

DID regression

- The DID regression, where `y81nrinc` = y81 × nearinc is the interaction term:

```r
options(scipen = 999)
reg_did <- lm(rprice ~ y81 + nearinc + y81nrinc, data = kielmc)
round(summary(reg_did)$coefficients, 4)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  82517.23   2726.910 30.2603   0.0000
y81          18790.29   4050.065  4.6395   0.0000
nearinc     -18824.37   4875.322 -3.8612   0.0001
y81nrinc    -11863.90   7456.646 -1.5911   0.1126
```

- The coefficient on `y81nrinc` matches the DID computed from the 2×2 table.
- The estimated DID effect is $\hat{\beta} = -\$11{,}864$ (SE = 7,457, $p = 0.113$). The incinerator reduced nearby house prices, but the effect is not statistically significant at the 5% level.

Assumptions in the incinerator example

- **Parallel trends:** without the incinerator, house prices near and far from the site would have followed the same trend over time.
- **No anticipation:** before the incinerator was announced, living near the future site did not affect house prices.

DID with covariates

- The basic DID estimator is unbiased only if the parallel trends assumption holds unconditionally. If houses near the incinerator site are systematically different from those farther away (e.g., older), and house age affects price trends, then near and far houses may follow different price trajectories even without the incinerator. This violates unconditional parallel trends.
- Omitting age from the regression biases the estimated common trend δ, which in turn biases the DID estimate β.
- Adding covariates addresses this under a weaker assumption: **parallel trends holds conditional on house age**. Among houses of the same age, near and far houses would have followed the same price trend absent the incinerator.

```
reg_did_cov <- lm(rprice ~ y81 + nearinc + y81nrinc + age + I(age^2),
                  data = kielmc)
round(summary(reg_did_cov)$coefficients, 4)
```

```
              Estimate Std. Error  t value Pr(>|t|)
(Intercept)  89116.5354  2406.0511  37.0385   0.0000
y81          21321.0418  3443.6311   6.1914   0.0000
nearinc       9397.9359  4812.2218   1.9529   0.0517
y81nrinc    -21920.2700  6359.7454  -3.4467   0.0006
age          -1494.4240   131.8603 -11.3334   0.0000
I(age^2)         8.6913     0.8481  10.2476   0.0000
```

- After controlling for house age, the DID estimate is $\widehat{\beta} = -\$21{,}920$ (SE $= 6{,}360$, $p = 0.0006$) — nearly twice as large and highly significant.
- The change in the estimate suggests that the basic DID was biased: omitting age biased the estimated trend, which in turn biased $\widehat{\beta}$.

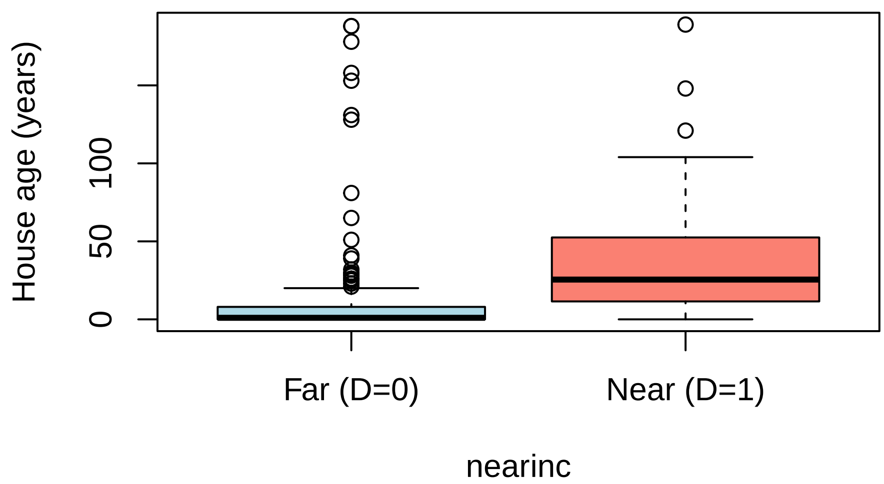
Where does the bias come from?

- Under no anticipation, the decomposition from the earlier slide gives:

$$\begin{aligned}\beta &= \text{ATT} \\ &\quad + \underbrace{\mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 1\right] - \mathrm{E}\left[Y_{i1}(0) - Y_{i0}(0) \mid D_i = 0\right]}_{\text{difference in trends}}.\end{aligned}$$

- **No anticipation is plausible:** the incinerator was not publicly known in 1978, so there is no reason for 1978 prices to already reflect the future incinerator.
- The bias must come from the difference in trends: a violation of **parallel trends**.
- Houses near the incinerator are systematically older:

```
boxplot(age ~ nearinc, data = kielmc,
        names = c("Far (D=0)", "Near (D=1)"),
        ylab = "House age (years)",
        col = c("lightblue", "salmon"))
```



- Since house age affects price trends nonlinearly, the near and far groups would have had **different price trends even without the incinerator**.
- The estimated bias is approximately $-\$11{,}864 - (-\$21{,}920) \approx +\$10{,}056$: the age-driven trend difference partially masked the incinerator's negative effect.

Assumption	Violated?	Reasoning
No anticipation	No	Incinerator not yet announced in 1978
Parallel trends	Yes (unconditionally)	Near houses are older; older houses had different price trends

Detecting anticipation and pre-trends

- With only two time periods (as in the incinerator example), anticipation effects **cannot** be separately identified.
- With **multiple pre-treatment periods**, an **event study** design can help. Suppose data span periods $t = -T, \ldots, -1, 0, 1, \ldots, T'$, with treatment at $t = 0$. Extend the DID regression by replacing the single interaction $\beta(t \cdot D_i)$ with a separate coefficient for each period:

$$
\begin{aligned}
Y_{it} = \alpha &+ \sum_{s \neq 0} \delta_s \cdot \mathbb{1}[t = s] + \gamma D_i \\
&+ \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.
\end{aligned}
$$

- Each δ_s captures the time effect in period s relative to $t = 0$ (analogous to δ in the two-period model). Each β_s measures how the treatment group's outcome differs from the control group in period s, beyond the baseline group difference γ and the common time effect δ_s.
- The pre-treatment coefficients β_s for $s < 0$ serve as a diagnostic:

Pattern in β_s for $s < 0$	Interpretation
All close to zero: $\beta_{-3} \approx \beta_{-2} \approx \beta_{-1} \approx 0$	Both assumptions look plausible
Sharp break right before treatment: $\beta_{-3} \approx \beta_{-2} \approx 0$ but $\beta_{-1} \neq 0$	**Anticipation**: units react to news of upcoming treatment
Gradual trend across pre-periods: $\lvert\beta_{-3}\rvert < \lvert\beta_{-2}\rvert < \lvert\beta_{-1}\rvert$, growing steadily	**Parallel trends violation**, not anticipation: groups were already diverging before treatment

Time fixed effects

- In the two-period model, the term $\delta \cdot t$ creates two intercepts: α at $t = 0$ and $\alpha + \delta$ at $t = 1$.
- With multiple periods $t = 0, 1, 2, \ldots$, we cannot use $\delta \cdot t$ because that forces a **linear** trend. Instead, include a separate dummy for each period. The regression actually estimated is:

$$\begin{aligned} Y_{it} = \alpha &+ \delta_1 \cdot \mathbb{1}[t = 1] + \delta_2 \cdot \mathbb{1}[t = 2] + \cdots \\ &+ \gamma D_i + \cdots + U_{it}. \end{aligned}$$

- The key property: $\mathbb{1}[t = s]$ equals 1 when $t = s$ and 0 otherwise. At $t = 0$ all dummies are zero (baseline). At any other t, exactly one dummy equals 1:

Period	Active dummy	Intercept
$t = 0$	none (baseline)	α
$t = 1$	$\mathbb{1}[t = 1] = 1$	$\alpha + \delta_1$
$t = 2$	$\mathbb{1}[t = 2] = 1$	$\alpha + \delta_2$

- Each period gets its own intercept. The compact notation $\sum_{s \neq 0} \delta_s \cdot \mathbb{1}[t = s]$ writes the time dummies as a sum. The event study model with all the dummies written out:

$$\begin{aligned} Y_{it} = \alpha &+ \delta_1 \mathbb{1}[t{=}1] + \delta_2 \mathbb{1}[t{=}2] + \cdots \\ &+ \gamma D_i + \beta_1 D_i \mathbb{1}[t{=}1] + \beta_2 D_i \mathbb{1}[t{=}2] + \cdots + U_{it}. \end{aligned}$$

- At each t, exactly one time dummy survives (and at $t = 0$ none do). So $\alpha + \delta_t$ is the intercept at time t. Define $\lambda_t = \alpha + \delta_t$ (with $\delta_0 = 0$):

$$\lambda_0 = \alpha, \quad \lambda_1 = \alpha + \delta_1, \quad \lambda_2 = \alpha + \delta_2, \quad \ldots$$

  The λ_t's are called **time fixed effects**. This is the regression we actually run in OLS, with a dummy for each period:

$$\begin{aligned} Y_{it} = \lambda_0 \mathbb{1}[t{=}0] &+ \lambda_1 \mathbb{1}[t{=}1] + \lambda_2 \mathbb{1}[t{=}2] + \cdots \\ &+ \gamma D_i + \beta_1 D_i \mathbb{1}[t{=}1] + \cdots + U_{it}. \end{aligned}$$

- At each t, exactly one λ-dummy equals 1 and all others are zero:

$$\underbrace{\lambda_0 \cdot 0 + \cdots + \lambda_t \cdot 1 + \cdots}_{\text{only } \lambda_t \text{ survives}} = \lambda_t.$$

  The notation λ_t is shorthand for this — it represents whichever λ is active at time t:

$$Y_{it} = \lambda_t + \gamma D_i + \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.$$

Individual fixed effects

- The same idea can be applied to individuals. In the event study model, the intercept for individual i is $\alpha + \gamma D_i$. This allows only **two** values:

Group	Intercept
Control ($D_i = 0$)	α
Treatment ($D_i = 1$)	$\alpha + \gamma$

  All individuals within the same group share the same baseline. But individuals may differ even within a group (e.g., houses near the incinerator differ in size, age, neighborhood quality).

- **Individual fixed effects** give each individual its own dummy, just like time fixed effects give each period its own dummy. The regression is:

$$
\begin{aligned}
Y_{it} = {} & \alpha_1 \mathbb{1}[i=1] + \cdots + \alpha_n \mathbb{1}[i=n] \\
& + (\text{time and treatment terms}) + U_{it}.
\end{aligned}
$$

  Only the dummy for individual i is active, so the intercept is α_i. Using time fixed effects λ_t from the previous slide:

$$
\begin{aligned}
Y_{it} = {} & \alpha_i + \lambda_t + \cancel{\gamma D_i} \\
& + \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.
\end{aligned}
$$

- Can we include both α_i and γD_i? **No.** D_i does **not change over time** for any individual, so it is already captured by α_i. Including both would be **perfect multicollinearity**:

$$
\begin{aligned}
Y_{it} = {} & \alpha_i + \lambda_t \\
& + \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it}.
\end{aligned}
$$

- The interaction terms $\beta_s \cdot D_i \cdot \mathbb{1}[t = s]$ survive because they vary across **both** individuals and time.

Two-way fixed effects

- Combining individual and time fixed effects gives the **two-way fixed effects** (TWFE) model:

$$
\begin{aligned}
Y_{it} = {} & \alpha_i + \lambda_t \\
& + \sum_{s \neq 0} \beta_s \cdot D_i \cdot \mathbb{1}[t = s] + U_{it},
\end{aligned}
$$

  where α_i is the individual fixed effect and λ_t is the time fixed effect.

- This is the standard specification for event studies and DID with multiple periods.